

Jiahong Li

Email: jiahong@caltech.edu
ORCID: 0000-0001-7938-9589

Education and Training

- Postdoc** in Medical Engineering Jun. 2026 -
- California Institute of Technology, Pasadena, CA
 - Advisor: Wei Gao
- Ph.D.** in Medical and Electrical Engineering Sep. 2020 - Jun. 2026
- California Institute of Technology, Pasadena, CA
 - Advisor: Wei Gao
- M.S.** in Materials Science and Engineering Sep. 2018 - Dec. 2019
- Northwestern University, Evanston, IL
 - Advisor: John Rogers
- B.S.** in Chemistry Sep. 2014 – Jun. 2018
- Wuhan University, Wuhan, China

Publication

- **Jiahong Li**, Kexin Fan, Xiaotian Ma, Canran Wang, Jihong Min, Yonglin Chen, Yu Song, Yadong Xu, Wei Gao*. “Strain-insensitive wet-tissue-adhesive biphasic bioelectronics for physicochemical monitoring and adaptive therapy” *Nature Materials* (2026) [\[Link\]](#)
- **Jiahong Li**, Jin Qu, Wei Gao*. “Tissue-bioelectronics interface.” *Chemical Society Reviews* (2026) [\[Link\]](#)
- Songsong Tang, Hong Han, Xiaotian Ma, Payal N Patel, Chen Gong, Junhang Zhang, Ernesto Criado-Hidalgo, Jounghyun Yoo, **Jiahong Li**, Gwangmook Kim, Shukun Yin, Di Wu, Mikhail G Shapiro, Qifa Zhou, Wei Gao*. “Enzymatic microbubble robots.” *Nature Nanotechnology* (2026) [\[Link\]](#)
- Elham Davoodi, **Jiahong Li**, Xiaotian Ma, Alireza Hasani Najafabadi, Jounghyun Yoo, Gengxi Lu, Ehsan Shirzaei Sani, Sunho Lee, Hossein Montazerian, Gwangmook Kim, Jason Williams, Jee Won Yang, Yushun Zeng, Lei S Li, Zhiyang Jin, Behnam Sadri, Shervin S Nia, Lihong V Wang, Tzung K Hsiai, Paul S Weiss, Qifa Zhou, Ali Khademhosseini, Di Wu, Mikhail G Shapiro, Wei Gao*. “Imaging-guided deep tissue in vivo sound printing.” *Science* (2025) [\[Link\]](#)
- Canran Wang†, Kexin Fan†, Ehsan Shirzaei Sani, José A Lasalde-Ramírez, Wenzheng Heng, Jihong Min, Samuel A Solomon, Minqiang Wang, **Jiahong Li**, Hong Han, Gwangmook Kim, Soyoung Shin, Alex Seder, Chia-Ding Shih, David G Armstrong, Wei Gao*. “A microfluidic wearable device for wound exudate management and analysis in human chronic wounds” *Science Translational Medicine* (2025) [\[Link\]](#)
- Hossein Montazerian†, Elham Davoodi†, Canran Wang, Farnaz Lorestani, **Jiahong Li**, Reihaneh Haghniaz, Rohan R Sampath, Neda Mohaghegh, Safoora Khosravi, Fatemeh Zehtabi, Yichao Zhao, Negar Hosseinzadeh, Tianhan Liu, Tzung K Hsiai, Alireza Hassani Najafabadi*, Robert Langer, Daniel G Anderson, Paul S Weiss*, Ali Khademhosseini*, Wei Gao*. “Boosting hydrogel conductivity via water-dispersible conducting polymers for injectable bioelectronics.” *Nature Communications* (2025) [\[Link\]](#)

- Wenzheng Heng, Shukun Yin, Jihong Min, Canran Wang, Hong Han, Ehsan Shirzaei Sani, **Jiahong Li**, Yu Song, Harry B Rossiter, Wei Gao*. "A smart mask for exhaled breath condensate harvesting and analysis" *Science* (2024) [\[Link\]](#)
- Yadong Xu, Zhilu Ye, Ganggang Zhao, Qihui Fei, Zehua Chen, **Jiahong Li**, Minye Yang, Yichong Ren, Benton Berigan, Yun Ling, Xiaoyan Qian, Lin Shi, Ilker Ozden, Jingwei Xie, Wei Gao*, Pai-Yen Chen*, Zheng Yan*. "Phase-separated porous nanocomposite with ultralow percolation threshold for wireless bioelectronics." *Nature Nanotechnology* (2024) [\[Link\]](#)
- Changhao Xu†, Yu Song†, Juliane R Sempionatto†, Samuel A Solomon†, You Yu, Hnin YY Nyein, Roland Yingjie Tay, **Jiahong Li**, Wenzheng Heng, Jihong Min, Alison Lao, Tzung K Hsiai, Jennifer A Sumner, Wei Gao*. "A physicochemical-sensing electronic skin for stress response monitoring." *Nature Electronics* (2024) [\[Link\]](#)
- **Jiahong Li**, Yadong Xu, Wei Gao* "Injectable and retrievable soft electronics." *Nature Materials* (2024) [\[Link\]](#)
- Cui Ye†, Minqiang Wang†, Jihong Min, Roland Yingjie Tay, Heather Lukas, Juliane R Sempionatto, **Jiahong Li**, Changhao Xu, Wei Gao*. "A wearable aptamer nanobiosensor for non-invasive female hormone monitoring." *Nature Nanotechnology* (2023) [\[Link\]](#)
- Yu Song†, Roland Yingjie Tay†, **Jiahong Li**, Changhao Xu, Jihong Min, Ehsan Shirzaei Sani, Gwangmook Kim, Wenzheng Heng, Inho Kim, Wei Gao*. "3D-printed epifluidic electronic skin for machine learning–powered multimodal health surveillance." *Science Advances* (2023) [\[Link\]](#)
- Jiaobing Tu, Jihong Min, Yu Song, Changhao Xu, Jiahong Li, Jeff Moore, Justin Hanson, Erin Hu, Tanyalak Parimon, Ting-Yu Wang, Elham Davoodi, Tsui-Fen Chou, Peter Chen, Jeffrey J. Hsu, Harry B Rossiter, Wei Gao*. "A wireless patch for the monitoring of C-reactive protein in sweat." *Nature Biomedical Engineering* (2023) [\[Link\]](#)
- Shirzaei Sani Ehsan†, Changhao Xu†, Canran Wang, Yu Song, Jihong Min, Jiaobing Tu, Samuel A. Solomon, **Jiahong Li**, Jaminelli L. Banks, David G. Armstrong, Wei Gao*. "A stretchable wireless wearable bioelectronic system for multiplexed monitoring and combination treatment of infected chronic wounds." *Science Advances* (2023) [\[Link\]](#)
- Yu You†, **Jiahong Li†**, Samuel A. Solomon†, Jihong Min, Jiaobing Tu, Wei Guo, Changhao Xu, Yu Song, and Wei Gao*. "All-printed soft human-machine interface for robotic physicochemical sensing." *Science robotics* (2022) [\[Link\]](#)
- Zhaoqing Cong, Songsong Tang*, Leiming Xie, Ming Yang, Yangyang Li, Dongdong Lu, **Jiahong Li**, Qingxin Yang, Qiwei Chen, Zhiqiang Zhang, Xueji Zhang, Song Wu*. "Magnetic-Powered Janus Cell Robots Loaded with Oncolytic Adenovirus for Active and Targeted Virotherapy of Bladder Cancer." *Advanced Materials* (2022) [\[Link\]](#)
- Xiaogang Guo†, Xiaoyue Ni†, **Jiahong Li†**, Hang Zhang, Fan Zhang, Huabin Yu, Jun Wu, Hongshuai Lei, Yonggang Huang, John A. Rogers*, Yihui Zhang*. "Designing mechanical metamaterials with kirigami-inspired, hierarchical constructions for giant positive and negative thermal expansion." *Advanced Materials* (2020) [\[Link\]](#)
- Xiaoyue Ni†, Xiaogang Guo†, **Jiahong Li**, Yonggang Huang, Yihui Zhang*, John A. Rogers*. "Two-Dimensional Mechanical Metamaterials with Widely-Tunable Unusual Modes of Thermal Expansion." *Advanced Materials*. (2019). [\[Link\]](#)
- Jinfeng Cao†, **Jiahong Li†**, Yumei Chen, Lina Zhang, and Jinping Zhou*. "Dual Physical Crosslinking Strategy to Construct Moldable Hydrogels with Ultrahigh Strength and Toughness." *Advanced Functional Materials* (2018) [\[Link\]](#)

- Sensors in Medicine. MedE 202, Caltech. Winter 2025.

Award and Honors

- Rising Star in Materials Science & Engineering, MIT, 2026
- Dr. David C. Gakenheimer Fellow, Caltech, 2024
- Andrew and Peggy Cherng Fellowship, Caltech, 2020
- Outstanding Graduation Thesis, Wuhan University 2018
- Samsung Fellowship, Wuhan University, 2016
- Outstanding Student Scholarship, Wuhan University, 2015